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The Targeting Rule Does Not Increase the Rate of Lower Extremity Injuries in NFL Players Over Two Seasons

Running Title: NFL Lower Extremity Injury Rate

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Abstract

Objectives:

In 2018 the NFL instituted playing rule article 8, which aimed to help decrease sport related concussion (SRC) by broadening the helmet-hit rule, making it a foul when a player lowers his head to make contact with his helmet against an opponent. Previous studies have demonstrated that the rate of lower extremity injuries increased following the institution of the targeting rule in NCAA football. The objective of this study was to determine if playing rule article 8

unintentionally led to a significant increase in the rate of lower extremity injuries sustained by NFL players during regular season games.

Methods: This study was a retrospective review of all NFL players who were placed on the publically available injury reports for a lower extremity injury from the 2016-2017 through 2019-2020 regular seasons. With article 8 taking effect starting with the 2018-2019 season, players were assigned to a pre- or post-rule cohort based on date of injury for purposes of injury rate comparison. Injury rates were calculated per 1000 athletic exposures (AE). Incidence rate ratio (RR) with 95% confidence intervals (CI) compared injury rates between the two cohorts.

Results:

There was a 3% decrease in the lower extremity injury rate in 2018-2019 to 2019-2020 compared with 2016-2017 to 2017-2018, however this was not statistically significant (33.6 versus 34.3/1000 AEs, respectively; RR, 0.97; 95% CI 0.92-1.04). There was a 40% decrease in the SRC rate when comparing the post- to the pre-article-8 cohort (3.3 vs 5.5/1000 AEs, respectively; RR 0.60; 95% CI 0.50-0.73).

Conclusions: The rate of lower extremity injuries and players placed on the injury reserve over the past two NFL seasons following the enactment of playing rule article 8 has not significantly increased. However, the average games missed due to lower extremity injuries significantly increased under playing rule article 8, which suggests that the severity of lower extremity injuries possibly increased.

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Level of evidence: IV Diagnostic

Introduction

Football has been a focal point of sports-related concussions (SRC) analysis and intervention. Given the increased awareness of the long-term consequences of multiple concussions, SRCs have gained increased societal interest over the past decade. In 1994 the National Football League (NFL) established the Committee on Traumatic Brain Injury, which has focused on reducing the incidence of SRCs [1]. More recently, the NFL Head, Neck and Spine Committee was formed to address neurological issues relevant to the health and safety of active NFL players. Similar to the NFL, the National Collegiate Athletic Association (NCAA) took steps to address the issue of SRC, and in 2008 the organization instituted a "targeting" rule to decrease the rate of SRC in football [2]. The 2008 NCAA "targeting" rule defined a foul as initiating contact with the crown of a helmet (Rule 9-1-3) or targeting a defenseless player in the head and neck area (Rule 9-1-4) punishable by personal foul penalty. In 2013 the NCAA expanded Rule 9-1-4, targeting a defenseless player in the head and neck area, with violation of the rule resulting in an automatic ejection from the game [3].

Interestingly lower extremity injury rates in NCAA football players were observed to increase following institution of the "targeting" rule over 6 seasons [2]. The increased incidence of injury was most pronounced in ankle and knee injuries, particularly those that were secondary to player contact [2]. However, the rates of noncontact/overuse injuries of the lower extremity did not change [2]. The authors speculate that despite the aim of the targeting rule change to decrease the incidence and severity of head trauma, it may have produced the unintended consequence of increased lower extremity injuries [2].

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In response to the NCAA targeting rule and increasing number of diagnosed concussions in NFL players, the NFL instituted playing rule article 8 prior to the 2018 season [4]. Playing rule article 8 aimed to help decrease SRC by broadening the helmet-hit rule, which now made it a foul when a player lowers his head to make contact with his helmet against an opponent [4]. The rule states that players found to be in violation of playing rule article 8 are to be penalized 15 yards and subject to possible disqualification from the game, pending replay review.

The aim of this study was to quantify the impact of playing rule article 8 on lower extremity injury rates among NFL players. Specifically, the objectives of this study were to determine if playing rule article 8 unintentionally led to a significant increase in (1) the rate of lower extremity injuries sustained by NFL players during regular season games, (2) the average games missed secondary to lower extremity injuries, and (3) the number of players placed on the injury reserve for lower extremity injuries.

When injury prevention interventions are instituted, such as rule changes, follow-up efficacy studies are recommended [5]. With growing awareness of player safety and the dangers associated with SRC, it is integral to quantify risk reduction in an objective manner. We hypothesized that rates of lower extremity injuries, average number of games missed, and percentage of players placed on IR would be significantly higher after the enactment of playing rule article 8 as compared to prior seasons.

Methods

Data collection

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This study was a retrospective review of all NFL players who were placed on the injury report for a lower extremity injury from the 2016-2017 through 2019-2020 NFL regular seasons. The data was compiled using publicly available injury reports (Appendix). The sources used publish weekly injury reports detailing the name of the players injured, their corresponding teams, and a brief description of each injury. At least two different sources were used to cross-reference injury information. To verify each player's date of injury, we confirmed each player's reported injury date corresponded to an absence of statistics from the respective games for which they were ruled out. These methods are in accordance with previously published protocols [6-10].

Team records available on individual team websites were used to collect demographic information. Our dataset included each player's name, position, team, pathology, age at injury, date of injury, season, number of weeks missed due to injury, and date of return to play.
With article 8 taking effect starting with the 2018-2019 season, players were assigned to a pre- or post-rule cohort based on date of injury for purposes of injury rate comparison.

Definitions

Reportable injuries were based on the criteria that the injury necessitated a team physician referral or emergent care and resulted in missed athletic activity. Pre-season games were excluded from investigation to account for the potentially different level of competition during this time period, as well as for the inability to track players lost to follow-up due to being cut from the team prior to the regular season. Additionally, post-season play was excluded so that all teams had equal representation throughout the study period to eliminate potential confounding

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of injury rates due to team-specific playing style. An athlete-exposure (AE) was defined as one athlete's participation in one game in which they were exposed to potential injury.

Data and Statistical Analysis

Data analysis was conducted using SPSS 25 (SPSS Inc, Chicago, Illinois). Injury rates were calculated per 1000 AEs using the following calculation: (Total number of lower extremity injuries (or SRC) x 1000/ Total number of AEs). Incidence rate ratio (RR) with 95% confidence intervals (CI) compared injury rates between the two cohorts. Average number of games missed and number of players placed on IR per year were also calculated. Continuous variables were statistically compared using two-tailed t-test, while categorical variables were compared using Chi-square test. Statistical significance was set at $P \leq 0.05$.

Results

Playing rule article 8 was enforced 45 times during its first two seasons accounting for 632 penalty yards.

Lower Extremity Injury Rates

During the 2016-2017 to 2019-2020 NFL seasons, 3685 lower extremity injuries were reported in regular season games, of which 1863 (50.5%) occurred during the 2016-2017 to 2017-2018 seasons and 1822 (49.5%) occurred during the 2018-2019 to 2019-2020 seasons (Table 1). There was a 3% decrease in the lower extremity injury rate in 2018-2019 to 2019-2020 compared with 2016-2017 to 2017-2018, however this was not statistically significant (33.6 versus 34.3/1000 AEs, respectively; RR, 0.97; 95% CI 0.92-1.04). Time loss (games) due to lower extremity injuries as well as number of players placed on the IR for lower extremity injuries was greater in the post playing rule article 8 cohort 2018-2019 to 2019-2020 seasons

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(5352 versus 4011 games loss due to LE injuries; 399 versus 346 players placed on the IR for LE injuries, respectively). There was a significant difference in time loss (games) due to LE injuries in 2018-2019 to 2019-2020 compared with 2016-2017 to 2017-2018.

Lower Extremity Injury Rates by Body Part

Most lower extremity injuries were to the knee (28.7%) and ankle (23.8%) and were caused by player contact (65.9%) (Table 2). There was no significant difference in injury rates in 2018-2019 to 2019-2020 compared with 2016-2017 to 2017-2018 for the knee (9.3 versus 10.1/1000 AEs, respectively, RR, 0.92; 95% CI 0.82-1.04) and ankle (7.8 versus 8.4/1000 AEs, respectively, RR, 0.93; 95% CI 0.81-1.06). Between the two cohorts there was no significant difference in the rate of lower extremity injuries because of player contact (22.3 versus 22.4/1000 AEs, RR, 0.99; 95% CI 0.91-1.07) or noncontact/overuse injuries (11.2 versus 11.8/1000 AEs, RR, 0.95; 95% CI 0.85 -1.06).

SRC Rates

Over the course of the study period of four NFL regular seasons, 479 SRCs were reported, of which 299 (62.4%) were reported prior to the targeting rule implementation and 180 (37.6%) after (Table 3). There was a 40% decrease in the SRC rate when comparing the post- to the pre-article-8 cohort (3.3 vs 5.5/1000 AEs, respectively; RR 0.60; 95% CI 0.50-0.73) (Table 4). Time loss due to SRC (games) was not significantly lower in the post playing rule article 8 cohort. There was no significant difference in percentage of players placed on the IR due to SRC between the two cohorts.

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Discussion

As rule changes are implemented, quantifying rule efficacy in promoting player safety should be vigorously evaluated. The purpose of this study was to determine if playing rule article 8 unintentionally led to a significant increase in (1) the rate of lower extremity injuries sustained by NFL players during regular season games, (2) the average games missed secondary to lower extremity injuries, and (3) the number of players placed on the injury reserve for lower extremity injuries. We hypothesized that rates of lower extremity injuries, average number of games missed, and percentage of players placed on IR would be significantly higher after the enactment of playing rule article 8 as compared to prior. Our findings suggest that over the past two seasons, the rate of lower extremity injuries and players placed on the IR for a lower extremity injury has not significantly increased, however the average games missed due to lower extremity injuries has significantly increased. Further the rate of SRC and average games missed due to SRC has significantly decreased, and players placed on the IR due to SRC has not significantly changed.

NFL players were first to voice concerns regarding the targeting rule; they speculated that the rule change may lead to an unintentional increase in the rate of lower extremity injuries as players, out of self-preservation and in an attempt to avoid head-on collisions, would opt to dive at the ball carriers' legs in order to make a tackle [11, 12]. Following this public criticism of the rule, in 2016 Westermann et al. investigated this claim and examined the rate of lower extremity injuries in NCAA football players following implementation of the targeting rule over six seasons. The findings of Westermann et al. suggested that lower extremity injury rates increased among NCAA football players after implementation of the targeting rule. Specifically, they noted that knee and ankle injuries resulting from player contact increased significantly [3]. The

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findings of Westermann et al. supported NFL players speculations that the targeting rule may increase the rate of lower extremity injuries. However, their study was not without limitation, to which they acknowledge and highlight potential confounders. The most significant confounder being the unintended effects of other policy changes made by the NCAA during the study period. For example in 2012 the NCAA moved the kickoff up to the 35-yd line (previously at the 30-yd line) in an attempt to decrease the rate of contact-related injuries. Although one can assume that this rule change would likely lead to more touchbacks, which in theory would reduce the risk of all contact related injuries. However, given these potential confounders further investigation in a cohort of NFL players is warranted.

As mentioned previously we did not find a significant difference in the rate of lower extremity injuries or players placed on the IR for a lower extremity injury over the past two NFL seasons under playing rule article 8. Our findings did note that the average games missed due to lower extremity injuries has significantly increased. This suggests that the severity of lower extremity injuries under playing rule article 8 has increased. However, this finding could also be secondary to a handful of players sustaining a season ending injury early in the season, and thus missing more games, which we did not control for in this study. The number of players placed on the IR for a lower extremity injury was higher under playing rule article 8 than the two previous seasons. However, the risk ratio of being placed on the IR for a lower extremity injury between our two cohorts was not significantly different. Despite this fact similar studies have demonstrated NFL rule changes to have an appreciable effect on the rate of lower extremity injuries in NFL athletes. Specifically, Baker et al. demonstrated that the NFL's chop block rule decreased the rate of lower extremity injuries in NFL defensive players [13]. Of note the chop

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block rule was instituted in 2015, thus it was not a confounding factor in this study. The findings of Baker et al. demonstrate that NFL rule changes can affect injury patterns in players.

The effect of the targeting rule on lower extremity injuries in NFL players can be explained by risk compensation theory. Risk compensation theory, first described by Hagel et al, suggests that people adjust their behavior according to the perceived severity of risk and may be less cautious with respect to less risky behavior [14]. As it relates to the current study, head-to-head contact and SRC during a play are the perceived risks, thus players may be adjusting their behavior, likely tackling form or body position upon contact, in order to avoid head-to-head contact. Following implementation of the targeting rule one can speculate that the perceived severity of risk of a head-to-head collision among football players is increased. The perceived increased risk prompts players to adjust their tackling form in order to avoid head-to-head collisions. Football players altering their tackling form, specifically tackling low in order to avoid head-to-head collisions, may increase the rate of lower extremity injuries in football players. This theory regarding the targeting rule increasing rates of lower extremity injuries in NCAA football players is supported by the findings of Westermann et al. In the current study, the observed rate of lower extremity injuries following the enactment of the targeting rule did not significantly increase. We speculate that this finding can be explained by a possible difference in perceived severity of risk of a head-to-head collision between collegiate and NFL football players. Our findings suggest that NFL players either do not perceive the risk of a head-to-head collision to be as severe as collegiate players and thus have not adjusted their tackling style to favor low tackles, or NFL players are, on average, more talented athletes and tackle with safer tackling form. However, neither of these speculations offer a good explanation of why players missed

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significantly more games due to LEI following institution of the targeting rule; this finding, though, may be related to timing of injury rather than severity, as mentioned previously.

This study provides important insight regarding patterns and injury rates in professional football players and can serve as a benchmark for discussion about injury prevention. To our knowledge, no other study has focused on rule article 8 and its effect on lower extremity injury rates in NFL players. Over the past century football has implemented rule changes, like the targeting and chop block rule, in order to improve player safety. Most notably in 1939 the National Collegiate Athletic Association (NCAA) made football helmets mandatory [2]. However, the most dramatic effect on player safety wasn't noted until the late 1970s when "spearing" was forbidden in order to improve the head position of defensive players when tackling. The "spearing" rule led to a significant decrease in the incidence of spinal cord injuries, demonstrating the powerful effect of rule changes on injuries in football [16].

Our study demonstrated that under playing rule article 8 the overall rate of SRC has decreased, while the average number of games missed and number of players placed on IR for a SRC did not significantly change. Similar to lower extremity injuries the average number of games missed and number of players placed on the IR reflects severity of injury. Thus, our study suggests that playing rule article 8 has significantly lowered concussion rate but not severity.

Since 2012 the NFL has publicly released injury data analyzed by IQVIA, an independent third-party company retained by the NFL [12]. The data published includes the incidence of reported concussions during the preseason and regular season. In comparing our data to the regular

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seasons data published by the NFL we noted that the NFL reported a higher rate of SRC for both the 2016-2017 to 2017-2018 and 2018-2019 to 2019-2020 cohorts (6.6 versus 5.5, and 5.1 versus 3.3 per 1,000 AEs; NFL compared to our data respectively). The data published by the NFL does also suggest that playing rule article 8 has decreased the rate of SRC over the two years it has been instituted, albeit less dramatically.

The discrepancy between our data and the data published by the NFL calls into question the validity of gathering publicly available data from injury reports to draw conclusions about injuries in professional athletes. However, this observation may in part be due to how the publishing of NFL injury reports work. It is important to note that NFL teams typically publish their first weekly injury report on Tuesday of each week, which is two days after a Sunday game but five days after a Thursday game. Thus, minor injuries sustained during NFL games that resolve during the 2-5-day period before the injury report is published would not be captured by the injury report, but may be captured by the NFL's own dataset. If twenty percent of the injuries reported by IQVIA resolved in the 2-5-day period before publication of team injury reports, it would fully explain the lower number of SRC published on injury reports.

This study, as well as other similar publications utilizing the well-established multistep protocol to identify injuries in professional athletes through review of public records, is associated with several well-recognized limitations. Details regarding injury diagnosis and management, including injury severity, exact pathology, imaging reports, and exact medical clearance were not available for all players. Because injured players were identified using public records, the possibility of reporting errors, and omissions exist. Study duration is a significant limitation as

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the study only spans four NFL seasons. Finally, there remains the possibility of selection bias in the injured group in which only "newsworthy" players injured were reported on. We attempted to control for this selection bias by only including regular season games, in order to account for the inability to track players lost to follow-up due to being cut from the team prior to the regular season. Further an attempt to minimize this bias was made by corroborating reported injuries with two additional resources, but the data is limited by what is publicly available.

Conclusion

The rate of lower extremity and players placed on the IR over the past two NFL seasons following the enactment of playing rule article 8 has not significantly increased. The average games missed due to lower extremity injuries has significantly increased under playing rule article 8, suggesting increased severity of injury. Further the rate of SRC and average games missed due to SRC has significantly decreased, and players placed on the IR due to SRC has not significantly changed.

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Appendix – Data Sources Utilized

http://www.nfl.com/injuries

https://www.espn.com/nfl/injuries

https://www.cbssports.com/nfl/injuries/

https://www.pro-football-reference.com/players/injuries.htm

https://bleacherreport.com/nfl-injuries

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Table 1. Lower Extremity Injuries in NFL Football Games by Season and Severity: 2016-2017 to 2019-2020 Seasons

	2016- 2018	2018-2020	Total	P-Value
All injuries	1863	1822	3685	0.5
Time loss due to LE injuries (games)	4011	5352	9363	0.01
No. of players placed on IR for LE injuries	346	399	745	0.06

*Lower extremity (LE), Injury reserve (IR)

Table 2. Lower Extremity Injuries in NFL Football Games by Season, Anatomic Site, and Injury Mechanism: 2016-2017 to 2019-2020 Seasons

	No. of Injuries (Injuries per 1000 AEs)			
Season and Anatomic Site	**Contact**	**Noncontact**	**Overuse**	**Total**
2016-2017 to 2017-2018				
Hip/groin	74 (1.3)	97 (1.8)	65 (1.2)	236 (4.3)
Upper leg/thigh	42 (0.7)	146 (2.7)	116 (2.1)	304 (5.6)
Knee	506 (9.3)	42 (0.7)	2 (0.03)	550 (10.1)
Lower leg/Achilles	30 (0.5)	67 (1.2)	39 (0.7)	136 (2.5)
Ankle	425 (7.8)	30 (0.5)	0	455 (8.4)
Foot/toes	144 (2.6)	22 (0.4)	16 (0.2)	182 (3.4)
Total	1221(22.4)	404 (7.4)	238 (4.4)	1863 (34.3)
2018-2019 to 2019-2020				
Hip/groin	93 (1.7)	192 (3.5)	64 (1.2)	349 (6.4)
Upper leg/thigh	21 (0.3)	149 (2.7)	48 (0.9)	218 (4.0)
Knee	499 (9.2)	10 (0.2)	0	509

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				(9.3)
Lower leg/Achilles	28 (0.5)	66 (1.2)	42 (0.7)	136 (2.5)
Ankle	410 (7.6)	14(0.3)	0	424 (7.8)
Foot/toes	160 (2.9)	26 (0.5)	0	186 (3.4)
Total	1211 (22.3)	457 (8.4)	154 (2.8)	1822 (33.6)
Overall				
Hip/groin	167 (1.5)	289 (2.7)	129 (1.2)	585 (5.4)
Upper leg/thigh	63 (0.6)	295 (2.7)	164 (1.5)	522 (4.8)
Knee	1005 (9.3)	52 (0.5)	2 (0.02)	1059 (9.7)
Lower leg/Achilles	58 (0.5)	133 (1.2)	81 (0.7)	272 (2.5)
Ankle	835 (7.8)	44 (0.4)	0	879 (8.1)
Foot/toes	304 (2.8)	48 (0.4)	16 (0.1)	368 (3.4)
Total	2432 (22.4)	861 (7.9)	392 (3.6)	3685 (33.9)

Table 3. SRC in NFL Football Games by Season and Severity: 2016-2017 to 2019-2020 Seasons

	2016- 2018	**2018-2020**	**Total**	**P-Value**
All SRC	299	180	479	**0.0001**
Time loss due to SRC (games)	367	264	631	0.61
No. of players placed on IR for SRC	16	11	27	0.33

*Sport related concussion (SRC), Injury Reserve (IR)

Table 4. Rate Ratio of SRC and LEI in NFL Football Games 2019-2020 Compared to 2016-2017 Seasons

	RR	**95% Confidence Interval**	**P-Value**
LEI per 1000 AE	0.97	0.92-1.04	0.5
Players placed on IR for LEI per	1.15	0.99-1.33	0.06

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1000 AE			
			P-Value
SRC per 1000 AE	**0.6**	**0.50-0.73**	**0.0001**
Players placed on IR for SRC per 1000 AE	0.68	0.31-1.48	0.33

*Sport related concussion (SRC), Lower extremity Injury (LEI), Athletic exposure (AE), Rate Ratio (RR)

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